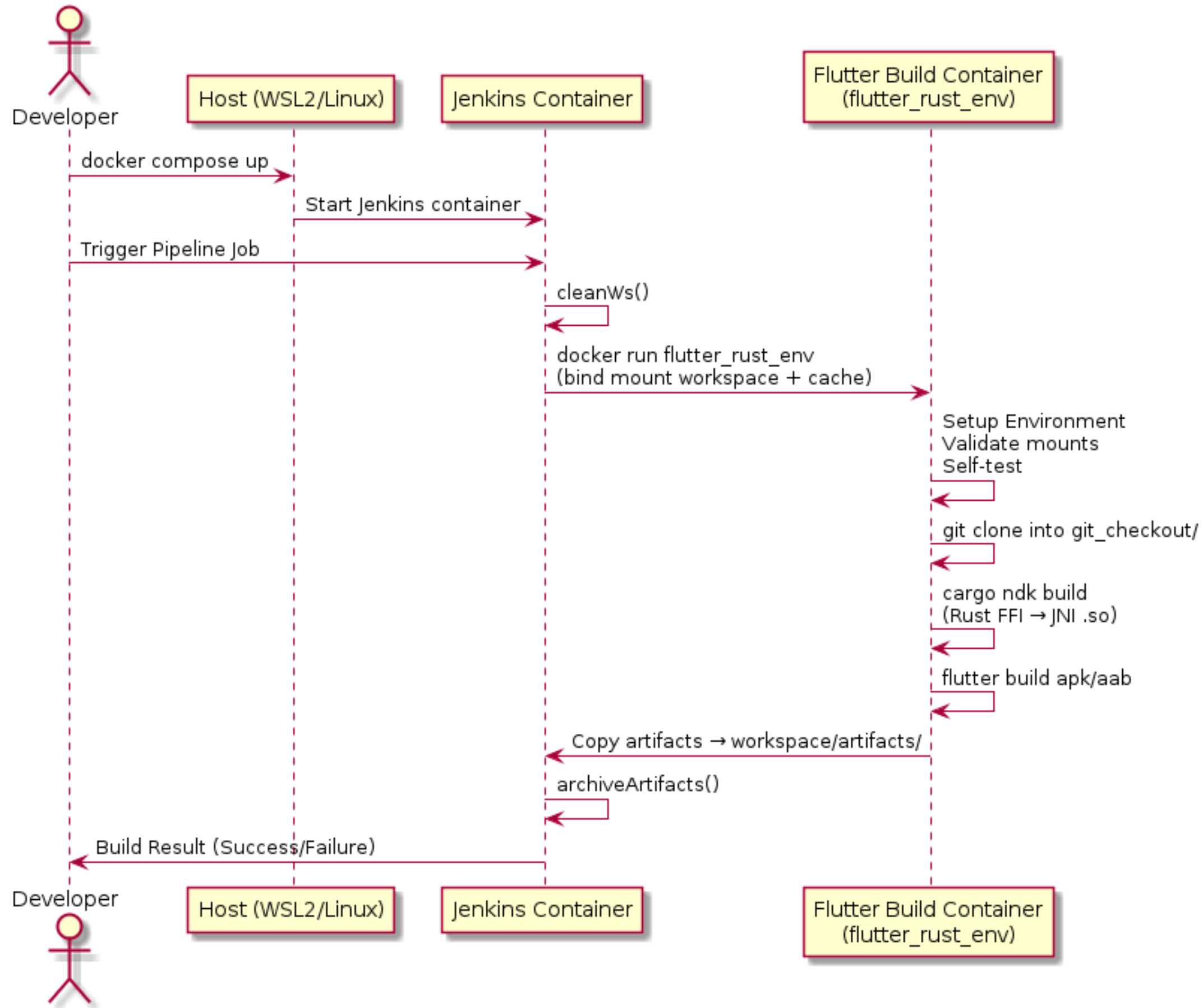
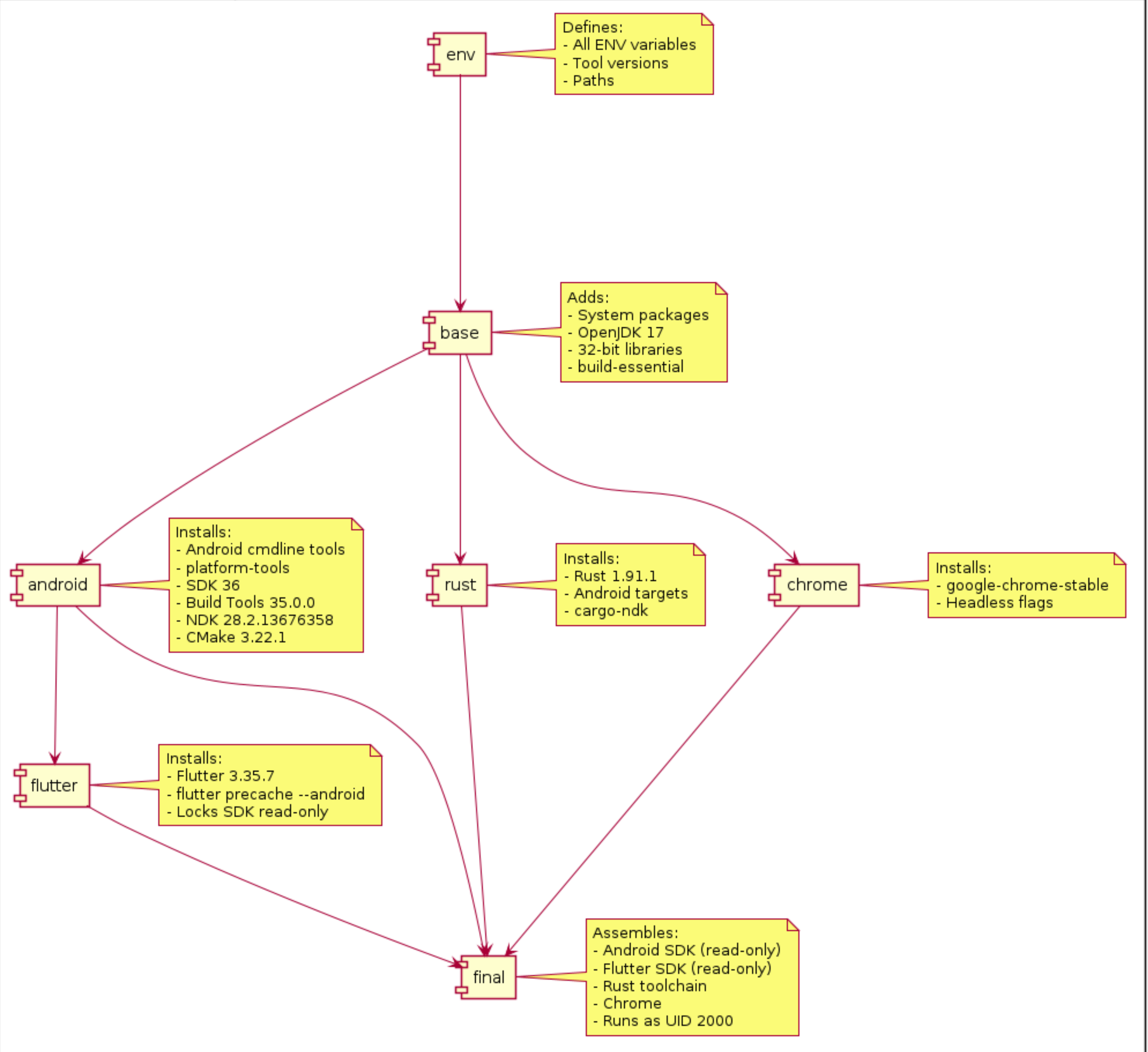


Deployment Sequence - Host → Jenkins → Flutter Container

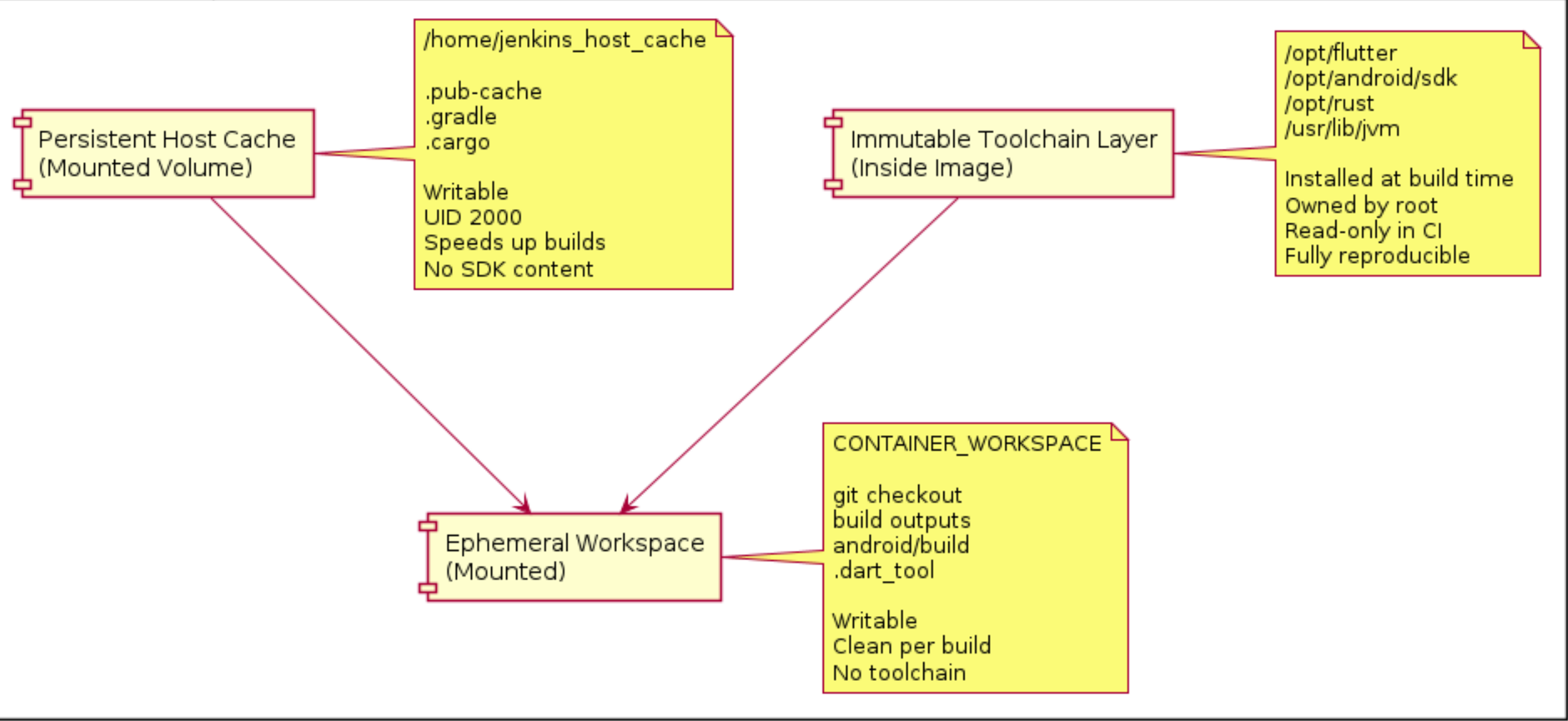


Immutable Toolchain Versions	
java	: 17
Flutter	: 3.35.7
Rust	: 1.91.1
Android SDK Tools	: 13114758
Android Compile	: 36
Android BuildTools	: 35.0.0
NDK	: 28.2.13676358
CMake	: 3.22.1
Base Image	: ubuntu 22.04

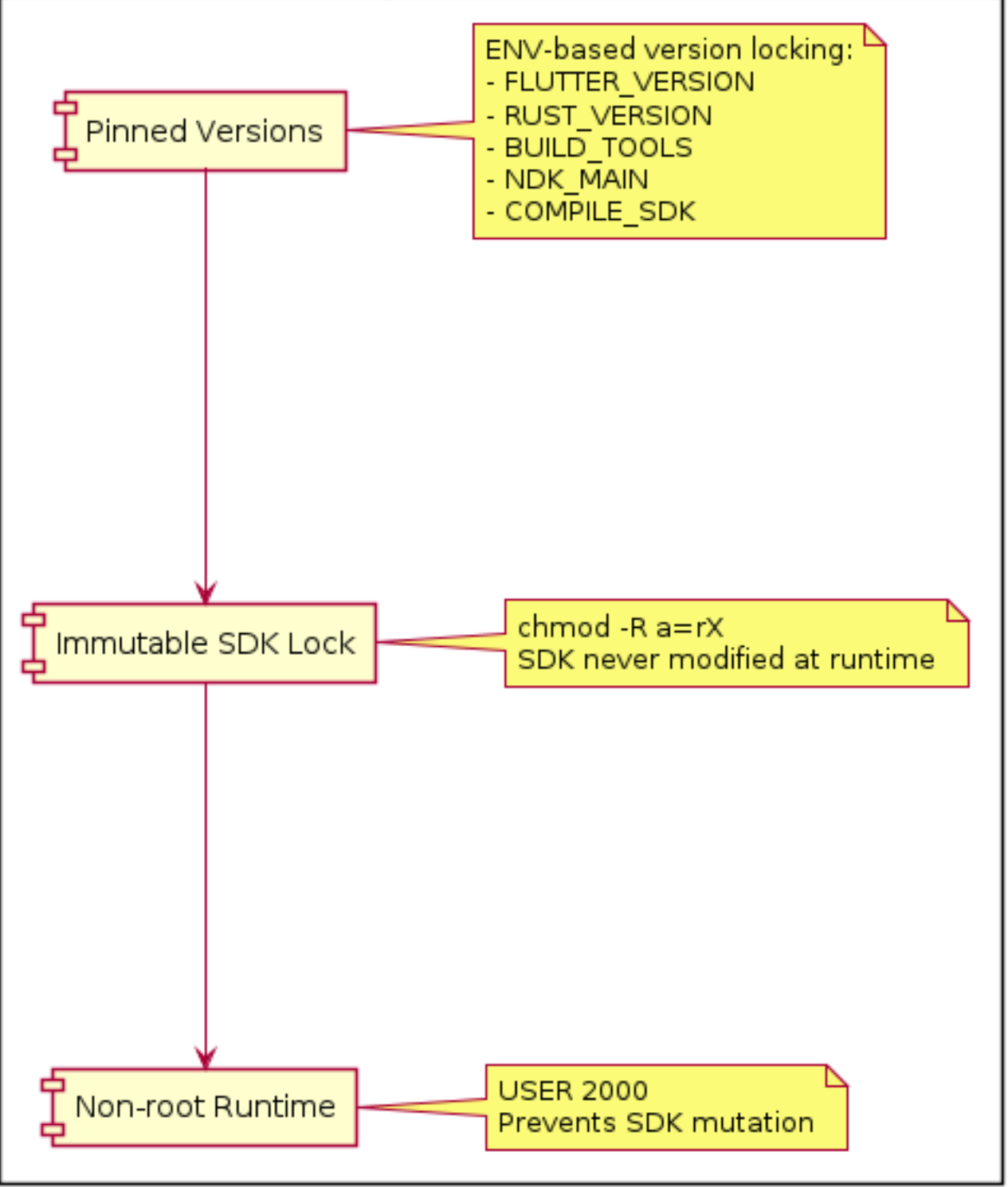
Docker Multi-Stage Build



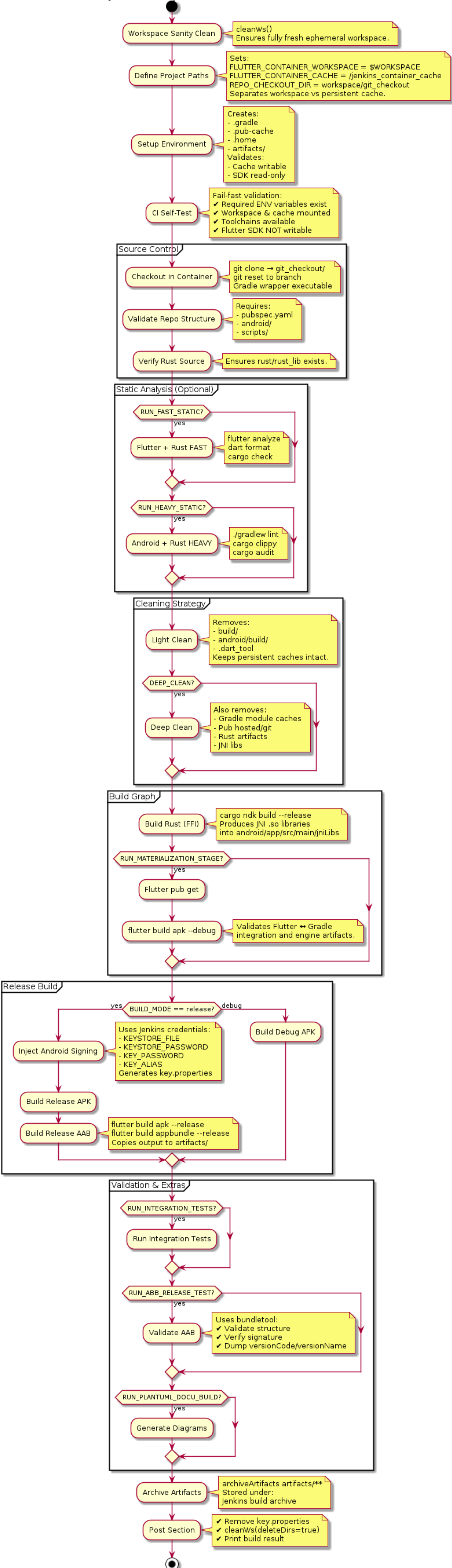
Security Model



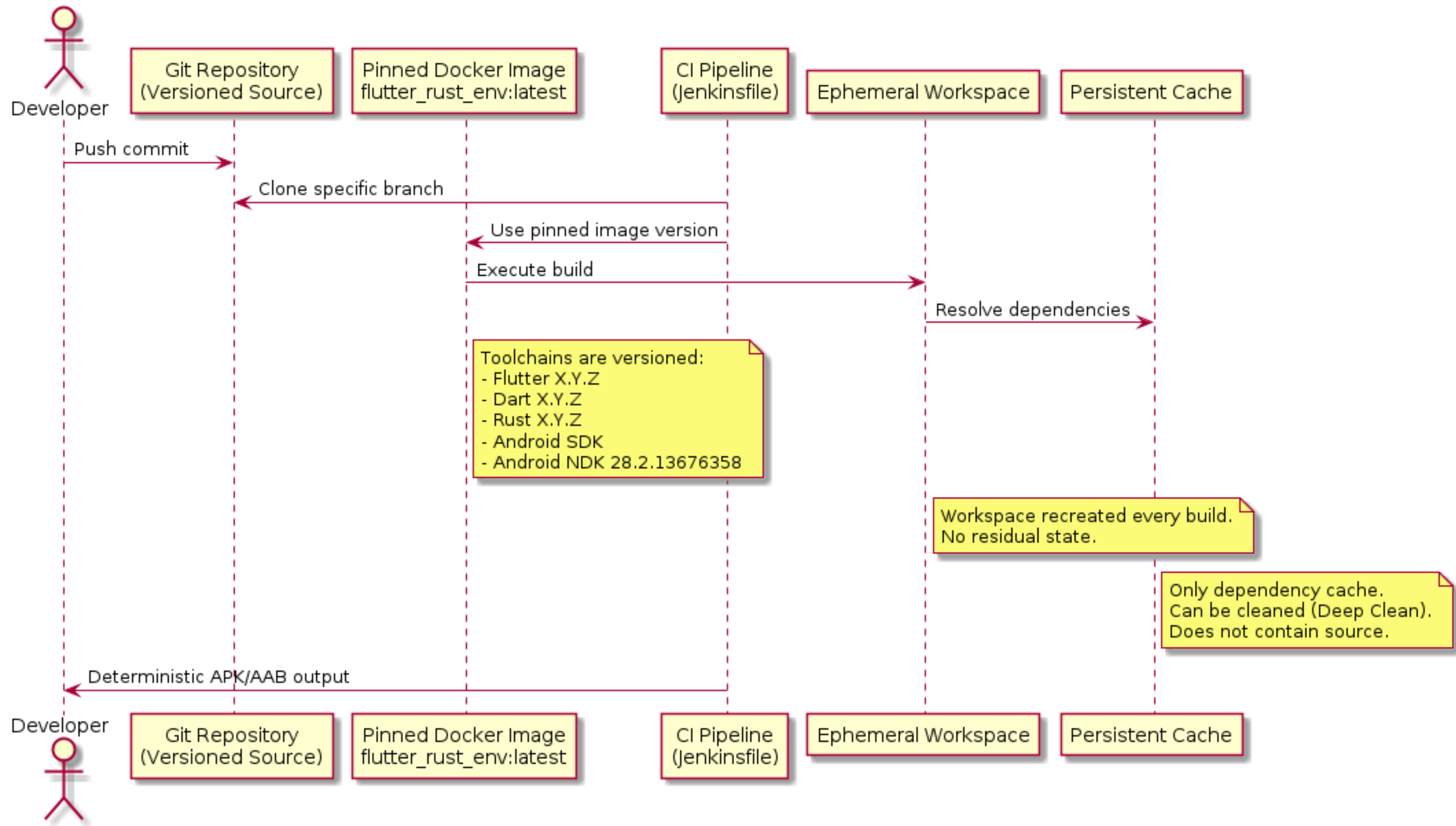
Reproducibility Model



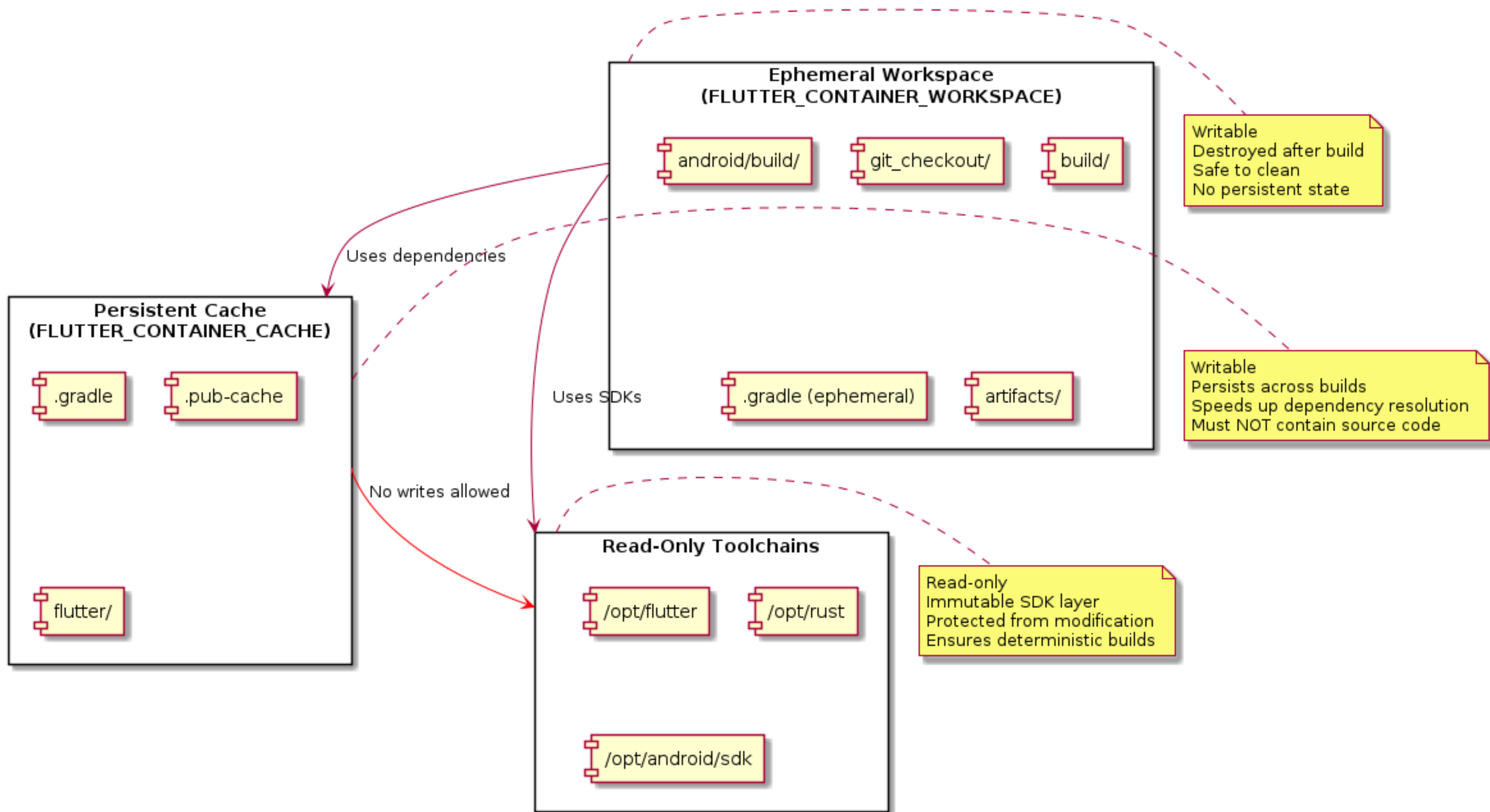
Jenkins CI Pipeline - Flutter + Rust + Android



Deterministic Build Model



Workspace vs Cache Security Model



My App Widgets

C _AddRemoveToggleState

List<bool> _selected
 DateTime? _lastPressed
 DateTime? _eraseAllConfirmedAt
 bool _eraseAllJustConfirmed
 DateTime? _savedGivensConfirmedAt
 bool _savedGivensJustConfirmed
 DateTime? _isResetToGivensConfirmedAt
 bool _isResetToGivensJustConfirmed
 DateTime? _isSelectAllCandConfirmedAt
 bool _isSelectAllCandJustConfirmed
 Color _iconColor
 build(BuildContext context): Widget
 buildElevatedToggle(int index): Widget
 _handleEraseAll(bool isDoubleTap): Future<void>
 _handleSaveGivens(bool isDoubleTap): Future<void>
 _handleResetToGivens(bool isDoubleTap): Future<void>
 _handleSelectAllCand(bool isDoubleTap): Future<void>

C AddRemoveToggle

AddRemoveToggle({key})
 createState(): _AddRemoveToggleState

Flutter Framework

C Color

C SnackBar

C ScaffoldMessenger

C Text

C Icon

C FittedBox

C ElevatedButton

C SizedBox

C Expanded

C Row

C Column

C Padding

C BuildContext

C Widget

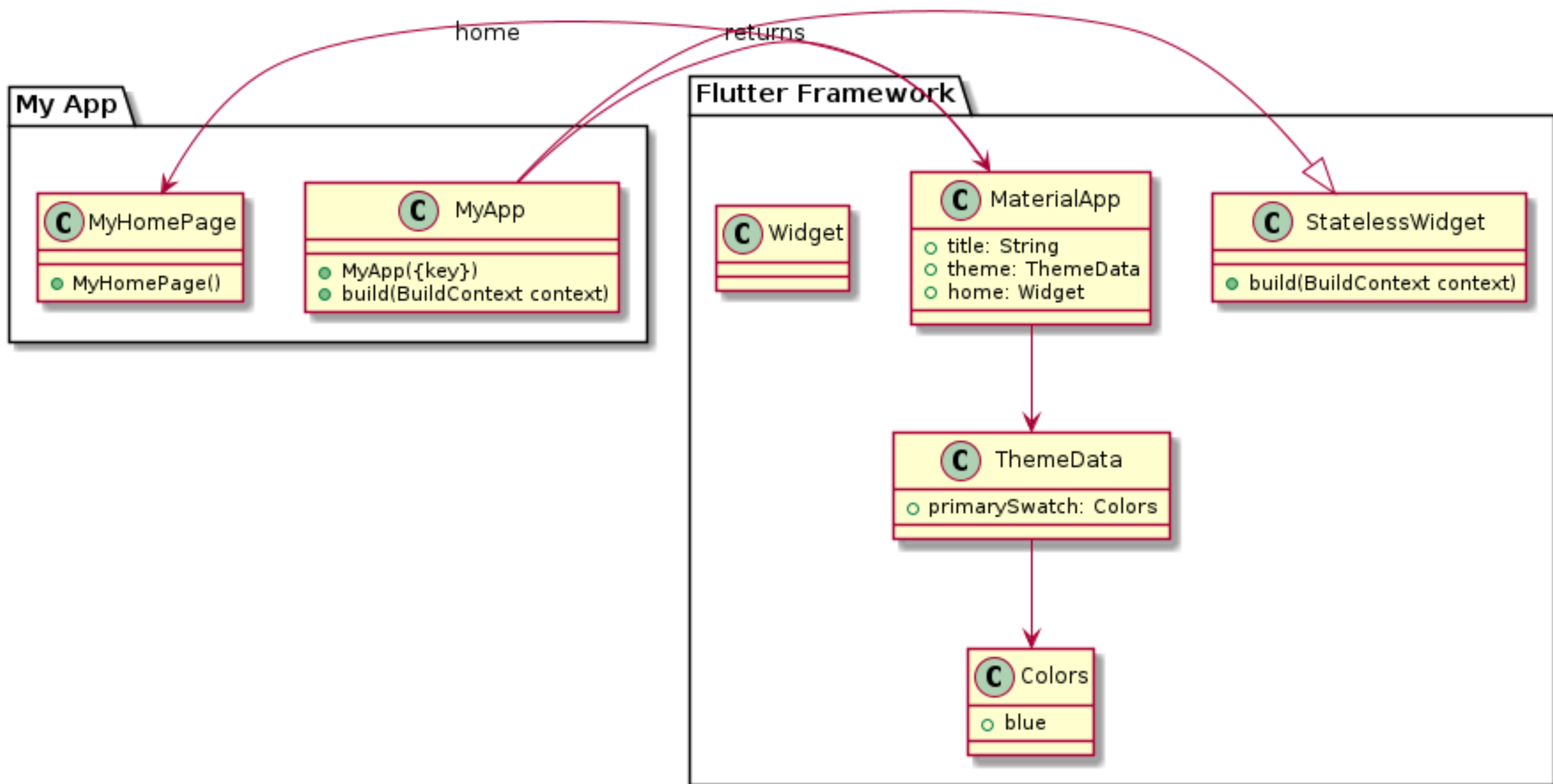
C State

StatefulWidget

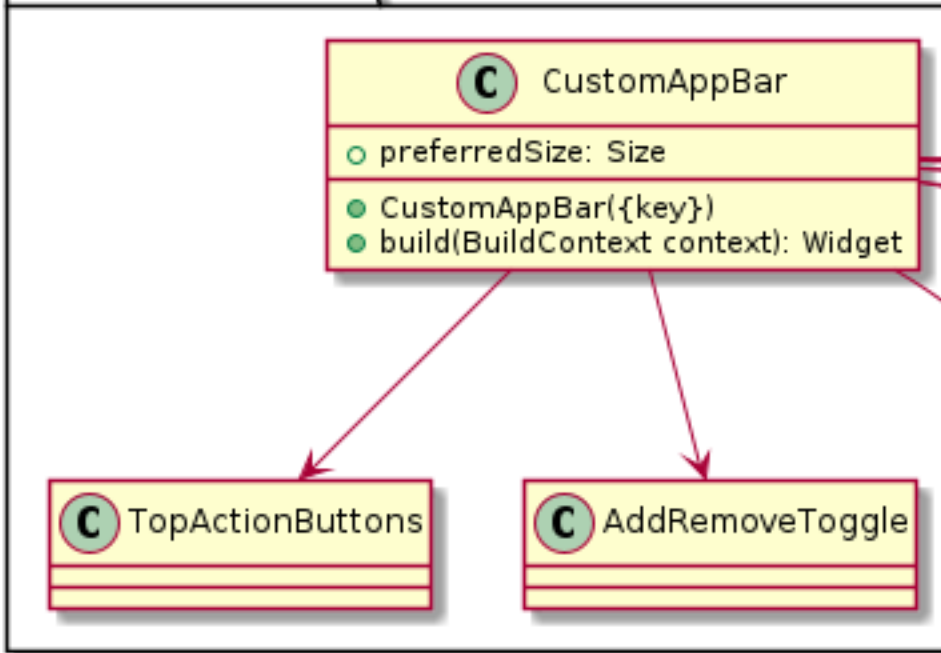
mounted: bool
 context: BuildContext
 build(BuildContext context): Widget
 setState(VoidCallback fn)

C StatefulWidget

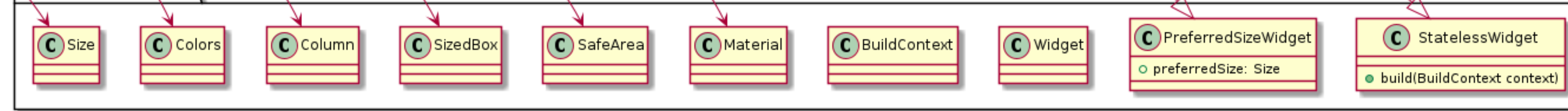
createState(): State



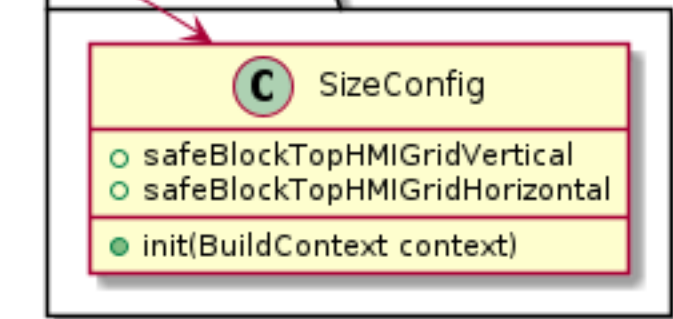
My App Widgets

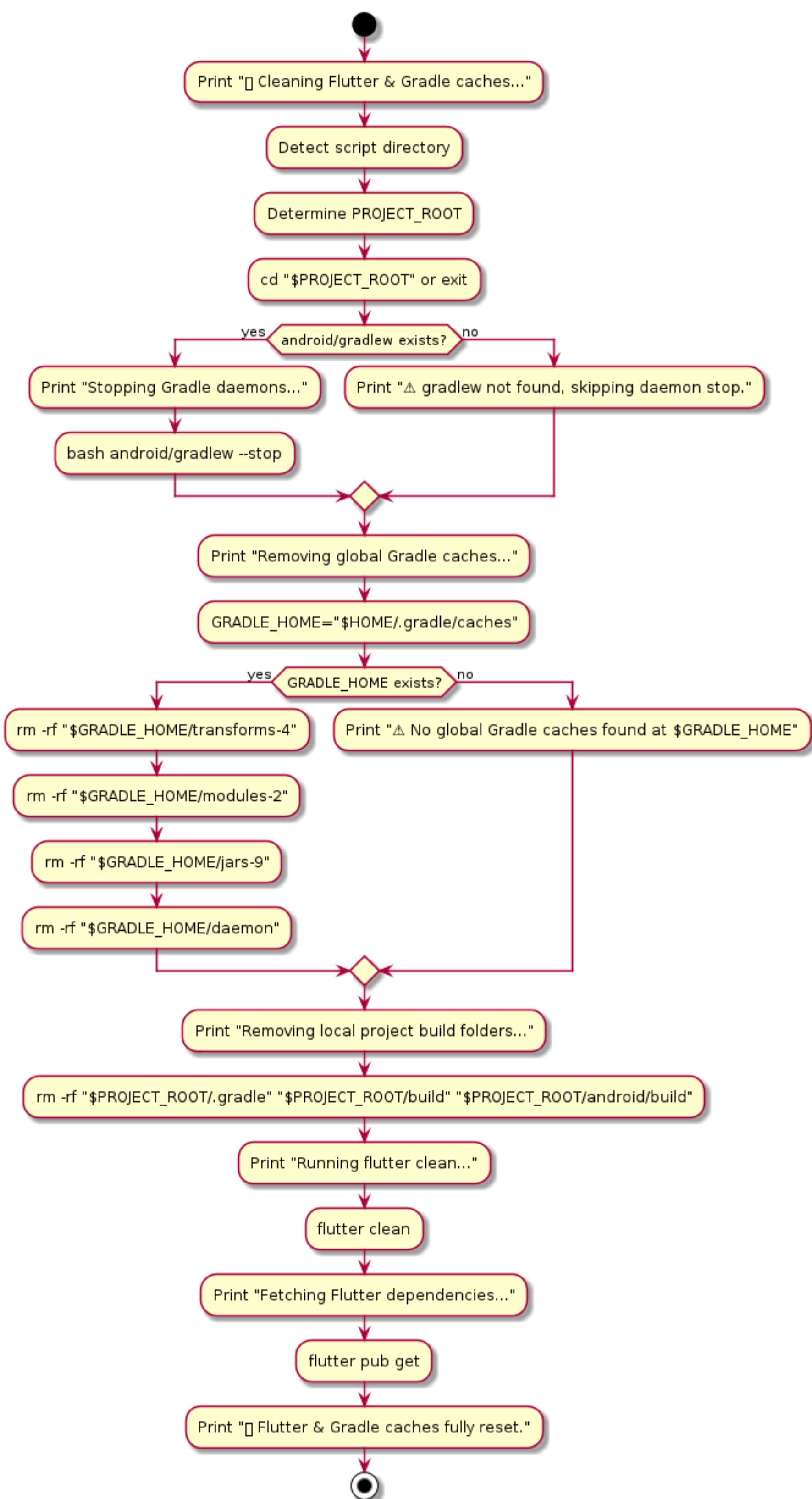


Flutter Framework

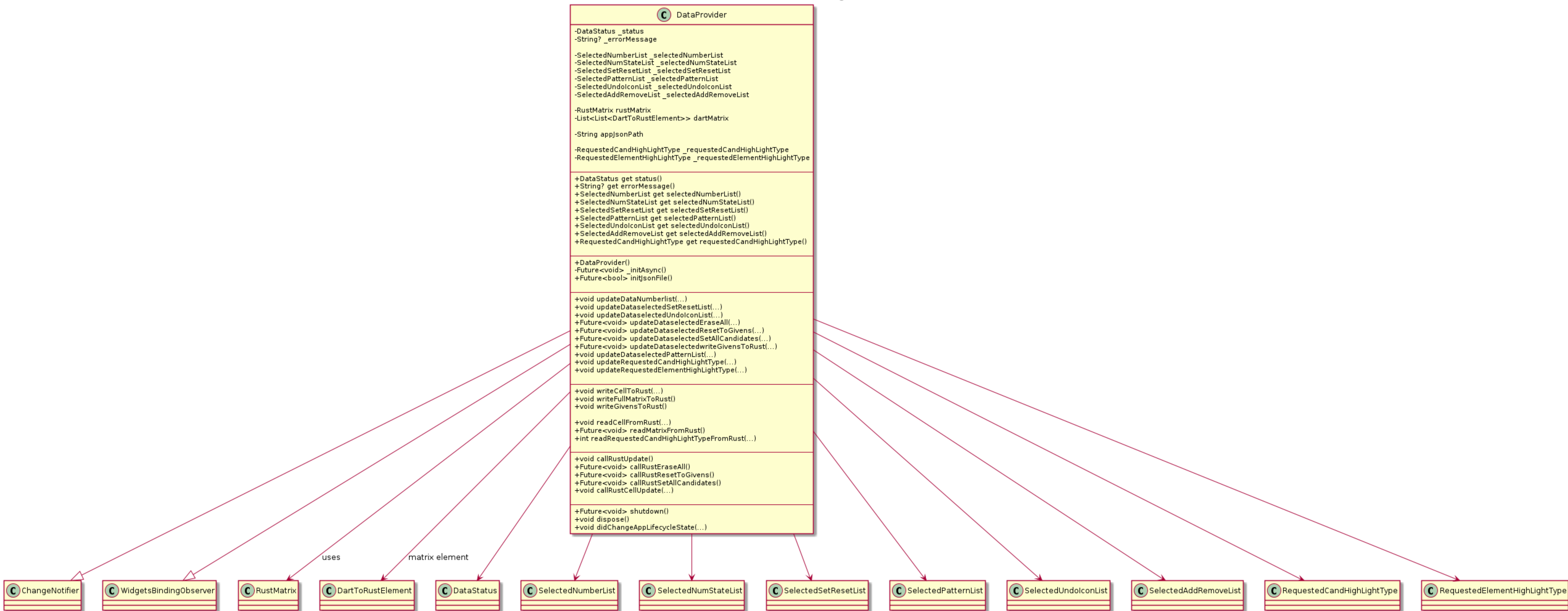


My App Utils





DataProvider Class Diagram



Export Barrel File - Module Dependency Diagram



C PatternList

- HI_LIGHT_ON: u8 = 0
- PAIRS: u8 = 1
- MATCH_PAIRS: u8 = 2
- TWINS: u8 = 3
- USER: u8 = 4

C Constants

- MAX_UINT: u8 = 255
- CONST_MATRIX_SIZE: u8 = 9
- CONST_MATRIX_ELEMENTS: u8 = 81
- constSelectedNumberListSize: u8 = 9
- constSelectedPatternListSize: u8 = 5
- constRequestedElementHighLightTypeSize: u8 = 5
- constRequestedCandHighLightTypeSize: u8 = 9
- constPatternListOff: u8 = 255

C MatrixFunctions

- create_matrix(rows: u8, cols: u8): *mut DartToRustElementFFI
- update_matrix(ptr: *mut DartToRustElementFFI, rows: u8, cols: u8): void
- free_matrix(ptr: *mut DartToRustElementFFI, rows: u8, cols: u8): void

Unsafe extern "C" function.
Allocates memory for DartToRustElementFFI array.
Returns null on allocation failure.

Unsafe extern "C" function.
Checks matrix elements and calls checkForElementPair.

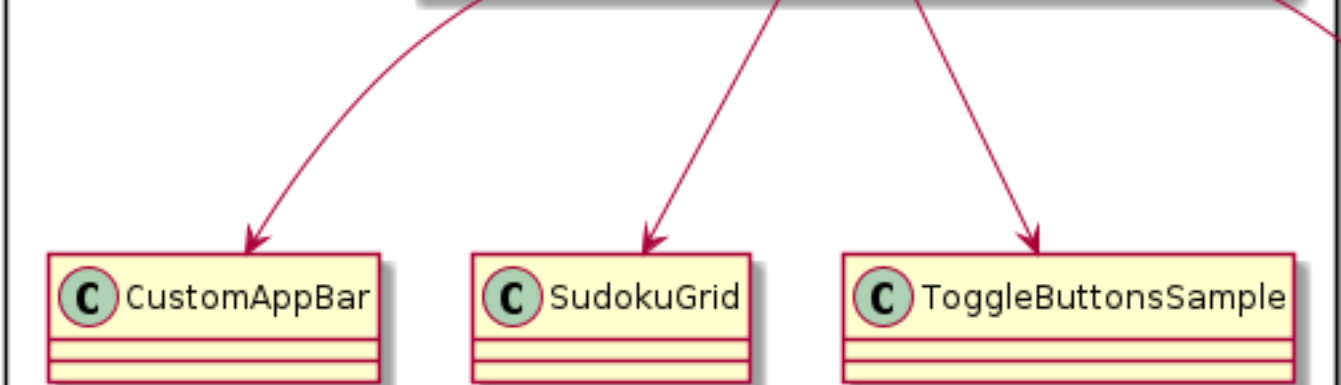
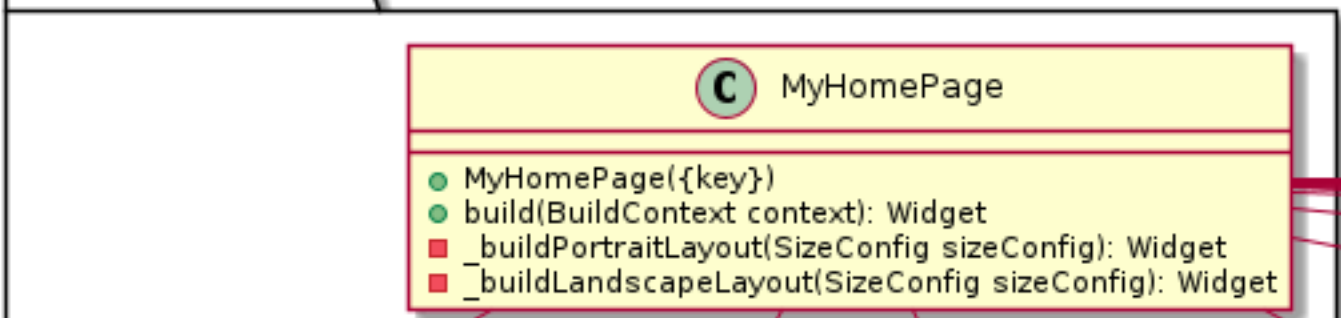
Unsafe extern "C" function.
Deallocates memory for DartToRustElementFFI array.

C DartToRustElementFFI

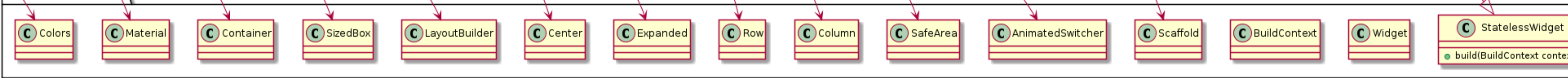
- row: u8
- col: u8
- selectedNum: u8
- selectedCandList: [u8; 9]
- selectedPatternList: [u8; 5]
- requestedElementHighLightType: [u8; 5]
- requestedCandHighLightType: [u8; 9]

1. [C Representation]
Used in FFI with Dart.
Unsafe operations wrapped in Rust functions.

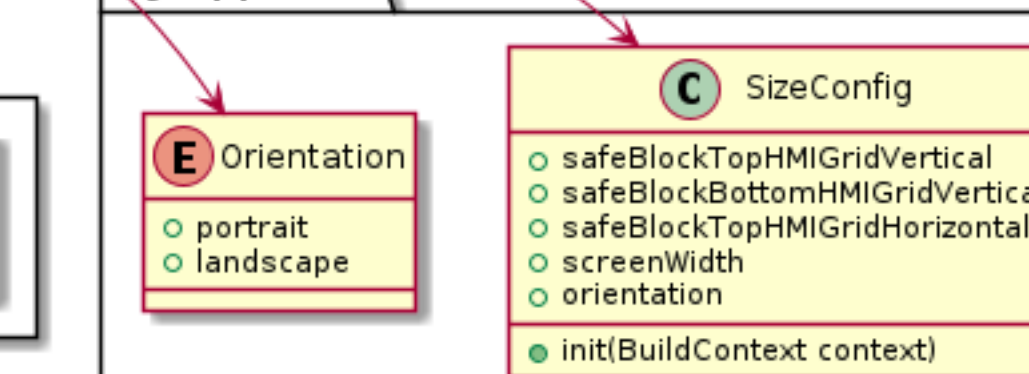
My App Widgets



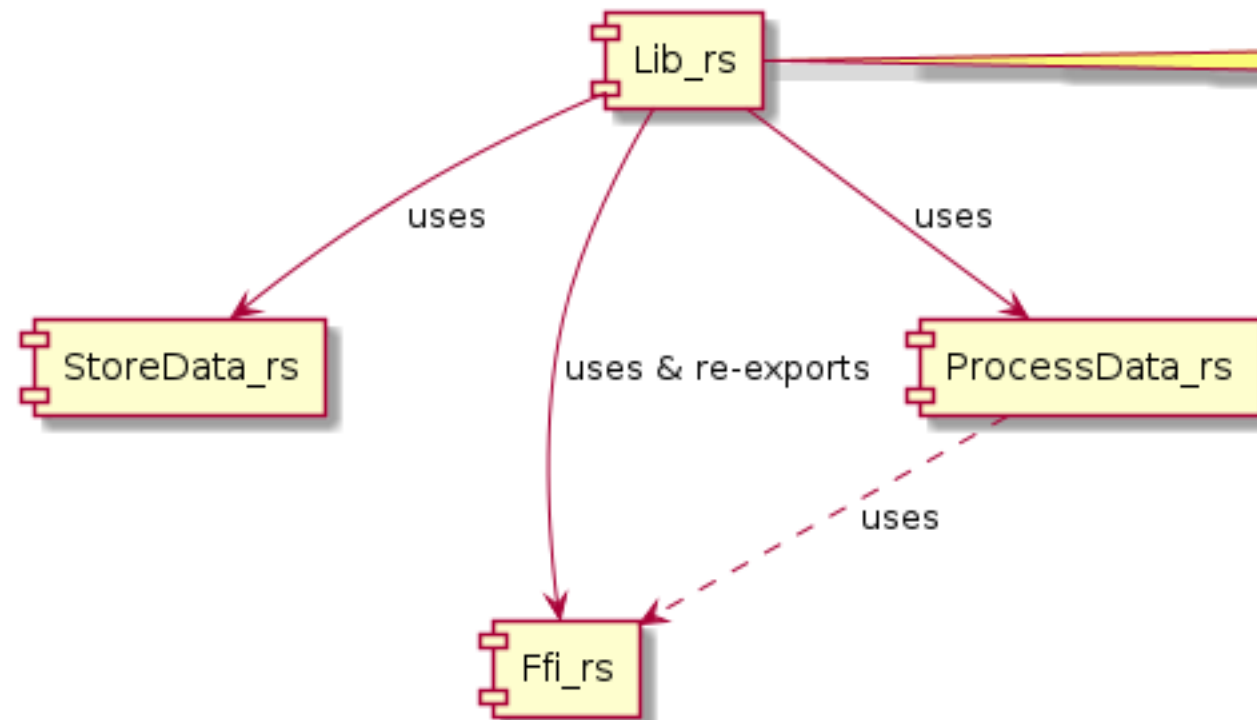
Flutter Framework



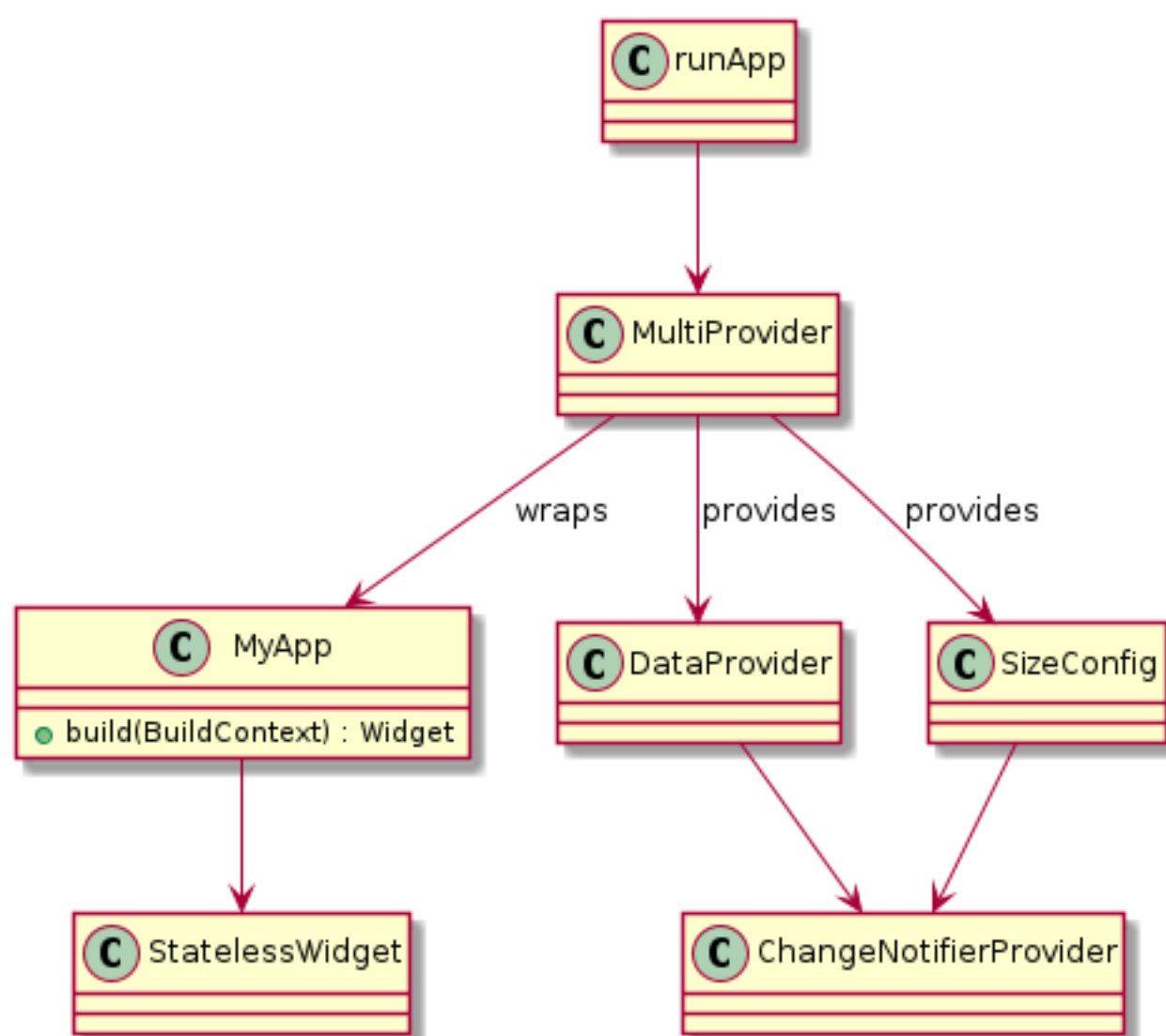
My App Utils

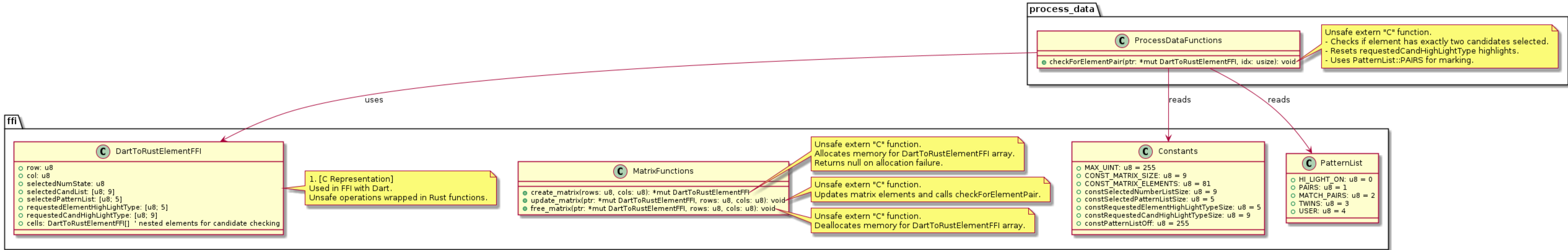


rust_lib_rs



This module contains the FFI bindings and is re-exported by lib.rs





Dart ↔ Rust FFI Architektur - Sudoku Solver

Dart Layer

C SudokuGrid

```
+List<List<int>> cells
+updateCell(int row, int col, int value)
+requestSolve()
+loadPuzzle(String path)
+savePuzzle(String path)
+RustMatrix nativeMatrix
```

owns

C RustMatrix

```
-Pointer<Void> ptr
+RustMatrix.fromSize(int size)
+setValue(int row, int col, int value)
+getValue(int row, int col) : int
+solve() : bool
+dispose()
```

uses via FFI

C FfiBindings

```
+Pointer<Void> matrix_create(int size)
+void matrix_free(Pointer<Void> ptr)
+void matrix_set(Pointer<Void> ptr, int r, int c, int v)
+int matrix_get(Pointer<Void> ptr, int r, int c)
+int matrix_solve(Pointer<Void> ptr)
+Pointer<Utf8> matrix_load(String path)
+void matrix_save(Pointer<Void> ptr, String path)
```

Rust Layer

C Error

```
+message : String
```

C Matrix

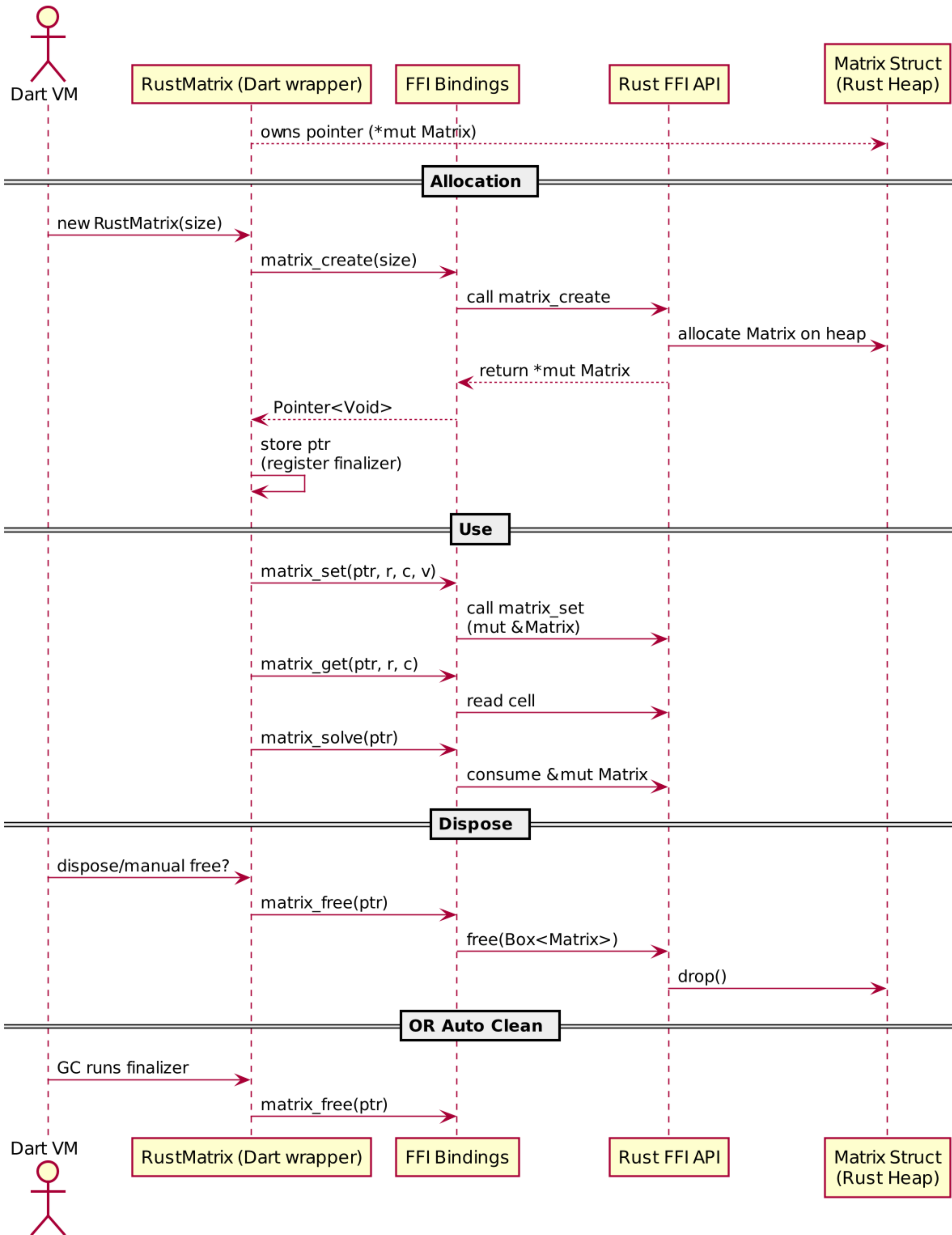
```
-Vec<Vec<u8>> cells
+new(size: usize)
+set(r: usize, c: usize, v: u8)
+get(r: usize, c: usize) : u8
+solve() : bool
+load(path: &str) : Result<Matrix, Error>
+save(path: &str) : Result<(), Error>
```

exposed via FFI

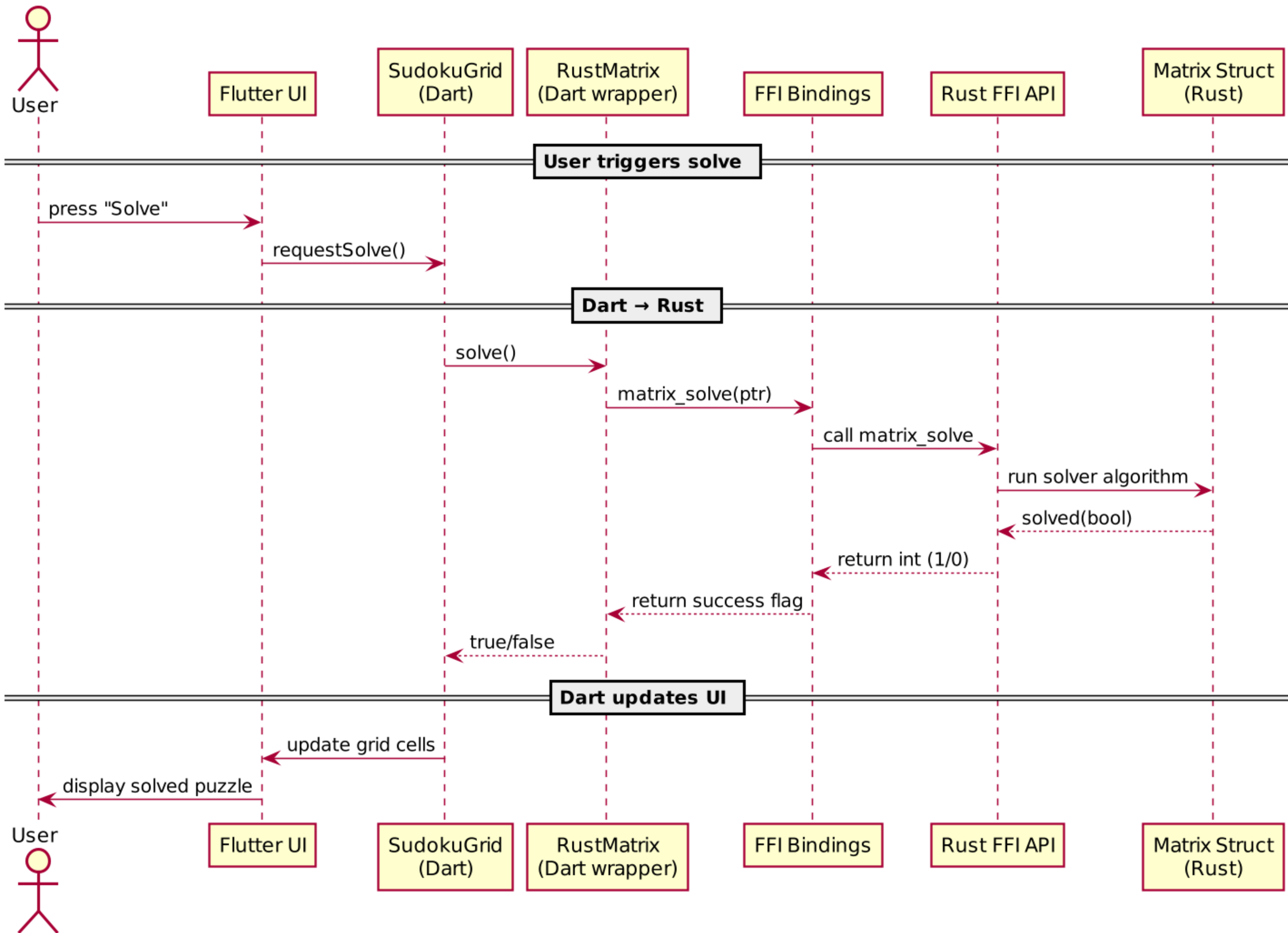
C FfiApi

```
+matrix_create(size: c_int) : *mut Matrix
+matrix_free(ptr: *mut Matrix)
+matrix_set(ptr: *mut Matrix, r: c_int, c: c_int, v: c_int)
+matrix_get(ptr: *mut Matrix, r: c_int, c: c_int) : c_int
+matrix_solve(ptr: *mut Matrix) : c_int
+matrix_load(path: *const c_char) : *mut Matrix
+matrix_save(ptr: *mut Matrix, path: *const c_char) : c_int
```

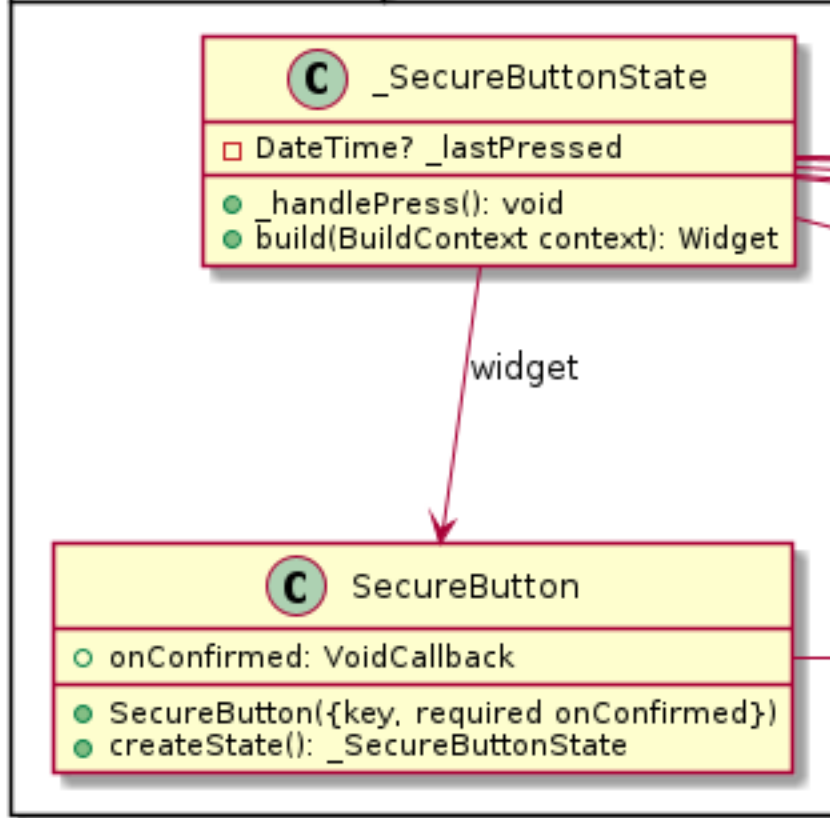
C - Pointer & Memory Lifecycle (Dart ↔ Rust FFI)



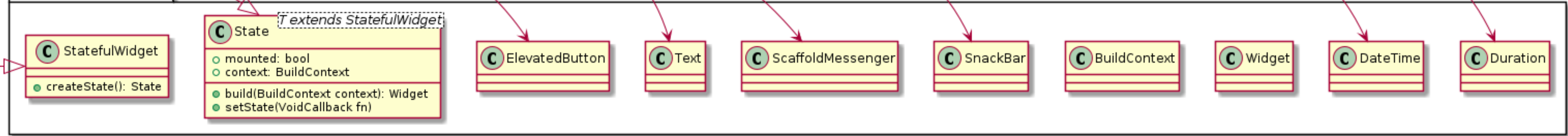
D - Sequenzdiagramm Solve-Ablauf (Dart ↔ Rust)

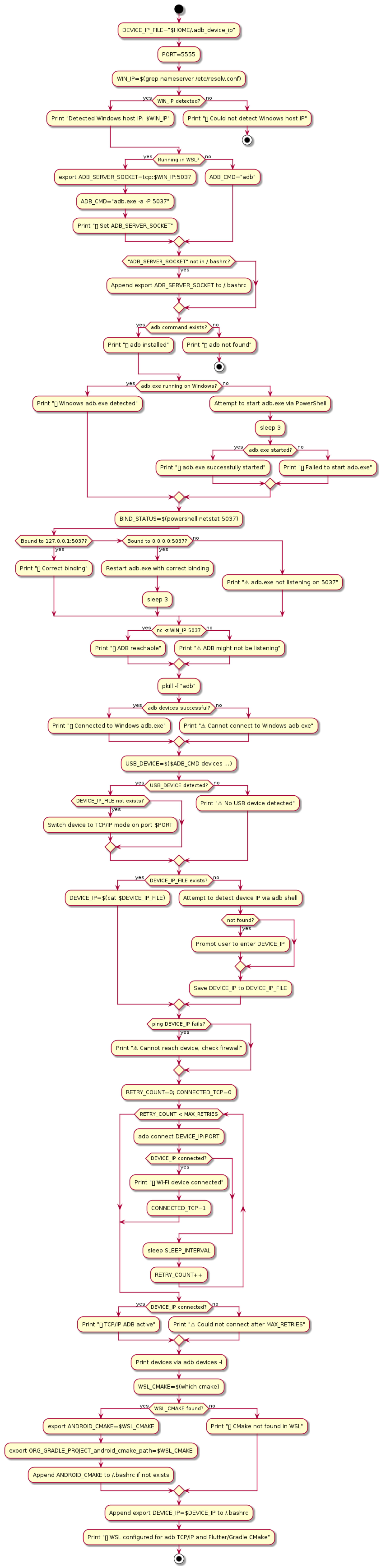


My App Widgets

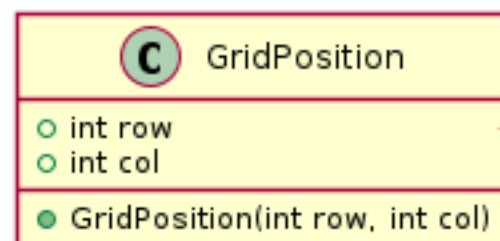


Flutter Framework

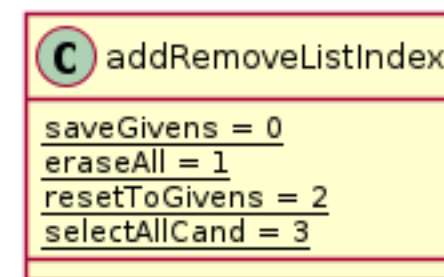
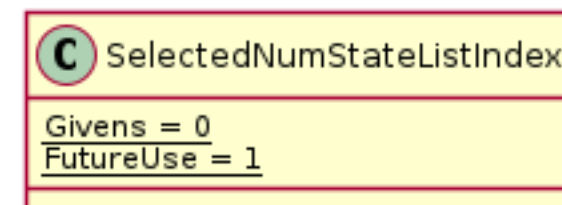
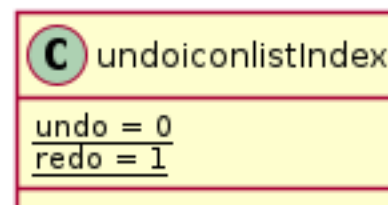
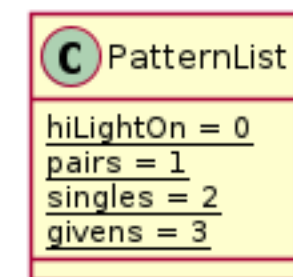
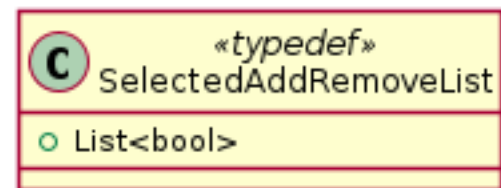
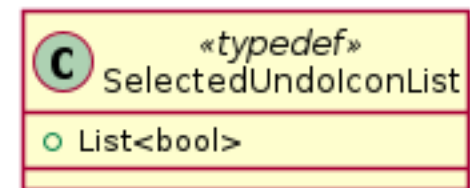
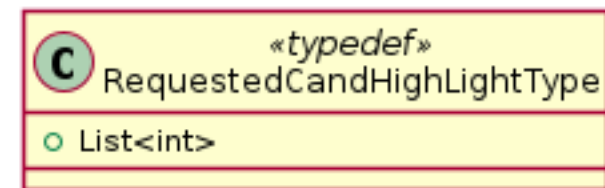
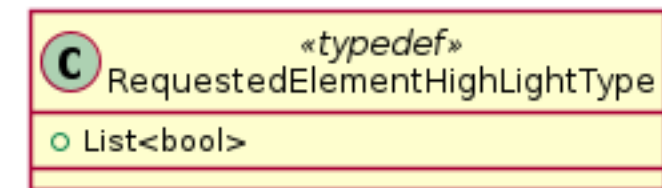
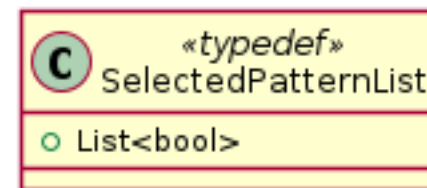
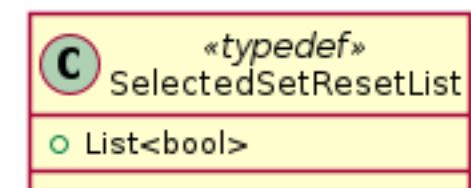
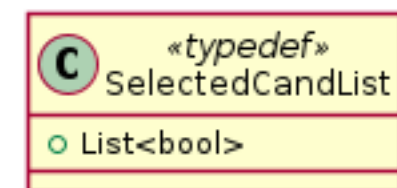
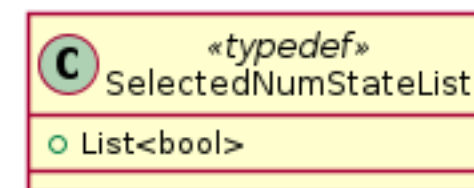
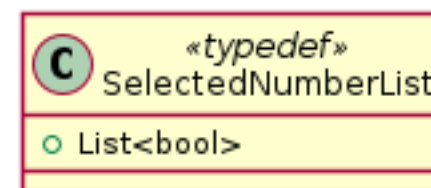
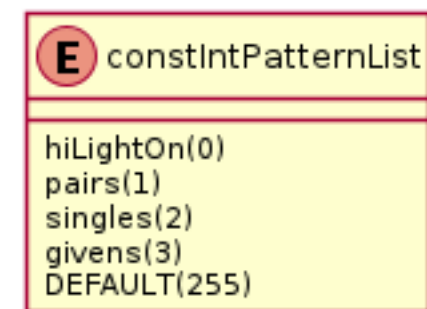
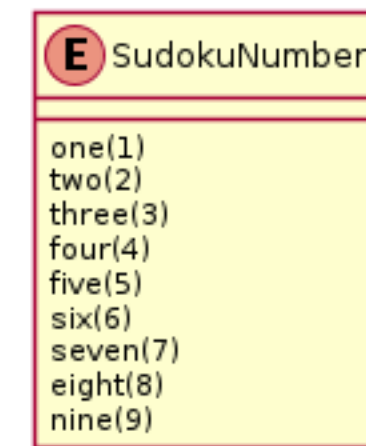
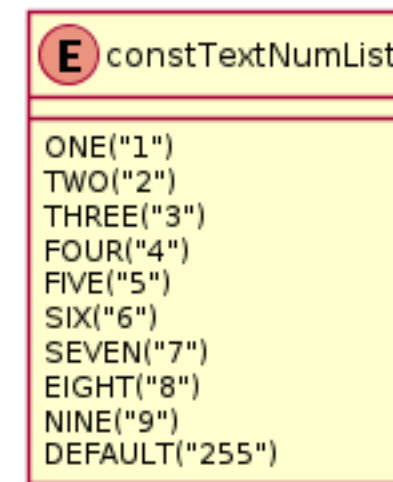
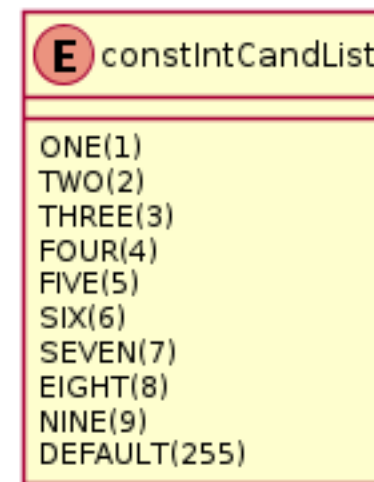
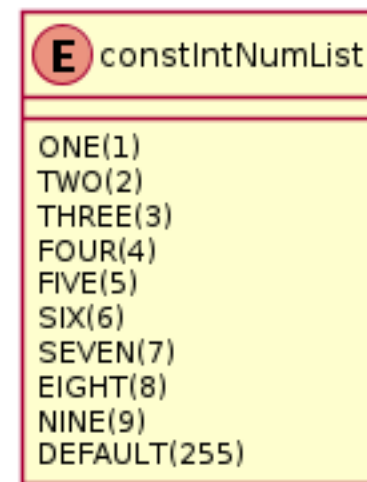
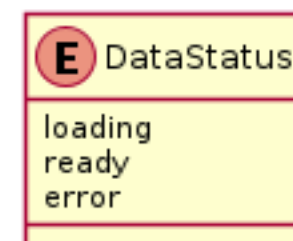




Shared Types, Enums, Constants, and Utility Classes



getRowColFromId(int id, int numColumns)
 - Computes global Sudoku coordinates
 - Based on 3×3 block subdivision



SizeConfig Class Diagram

C SizeConfig

Fields

- MediaQueryData _mediaQueryData
- Orientation orientation
- double screenWidth
- double screenHeight
- double blockSizeHorizontal
- double blockSizeVertical
- double safeAreaHorizontal
- double safeAreaVertical
- double safeBlockHorizontal
- double safeBlockVertical

- double safeBlockTopHMIGridVertical
- double safeBlockMidSudokuGridVertical
- double safeBlockBottomHMIGridVertical

- double safeBlockTopHMIGridHorizontal
- double safeBlockMidSudokuGridHorizontal
- double safeBlockBottomHMIGridHorizontal

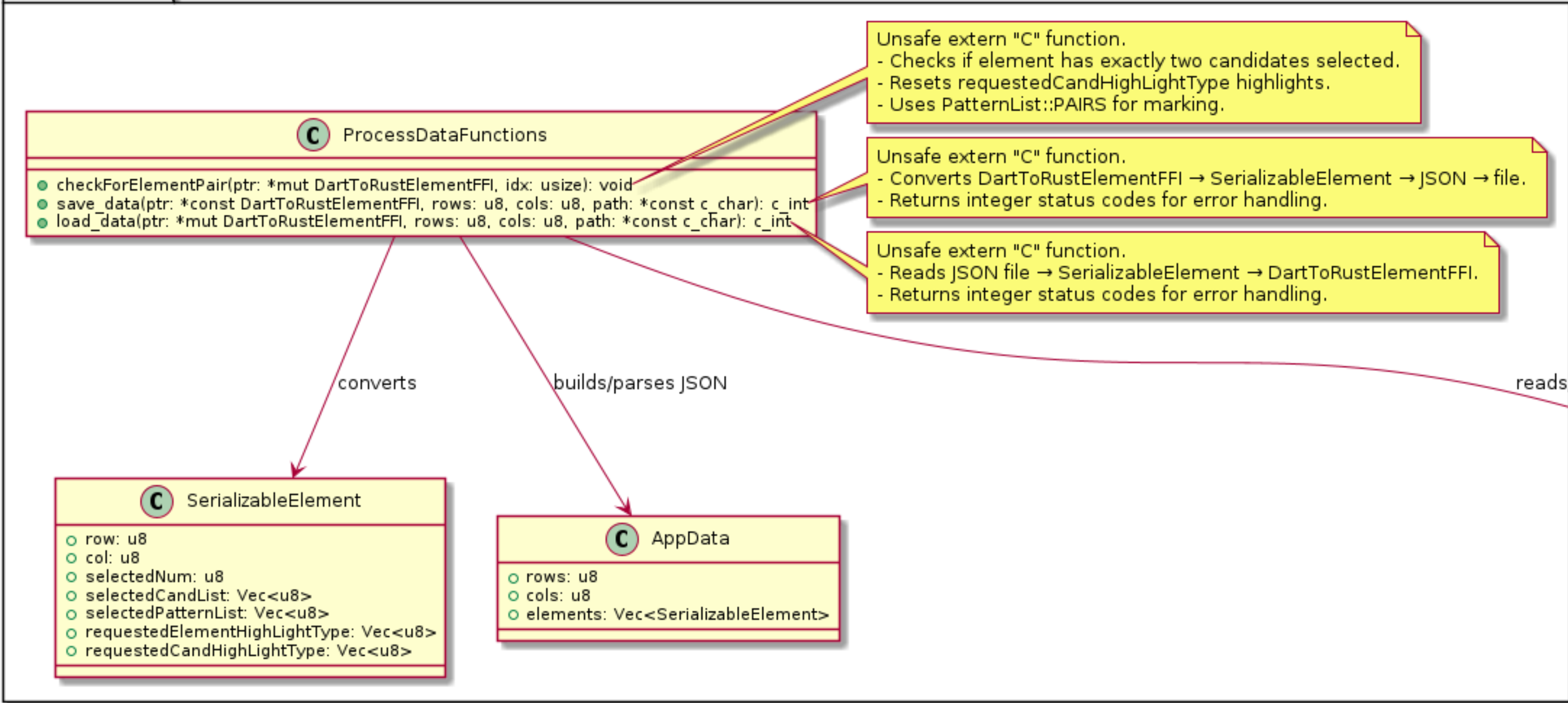
Methods

- void init(BuildContext context)
- void _calculatePortrait()
- void _calculateLandscape()
- void _validateLayout()

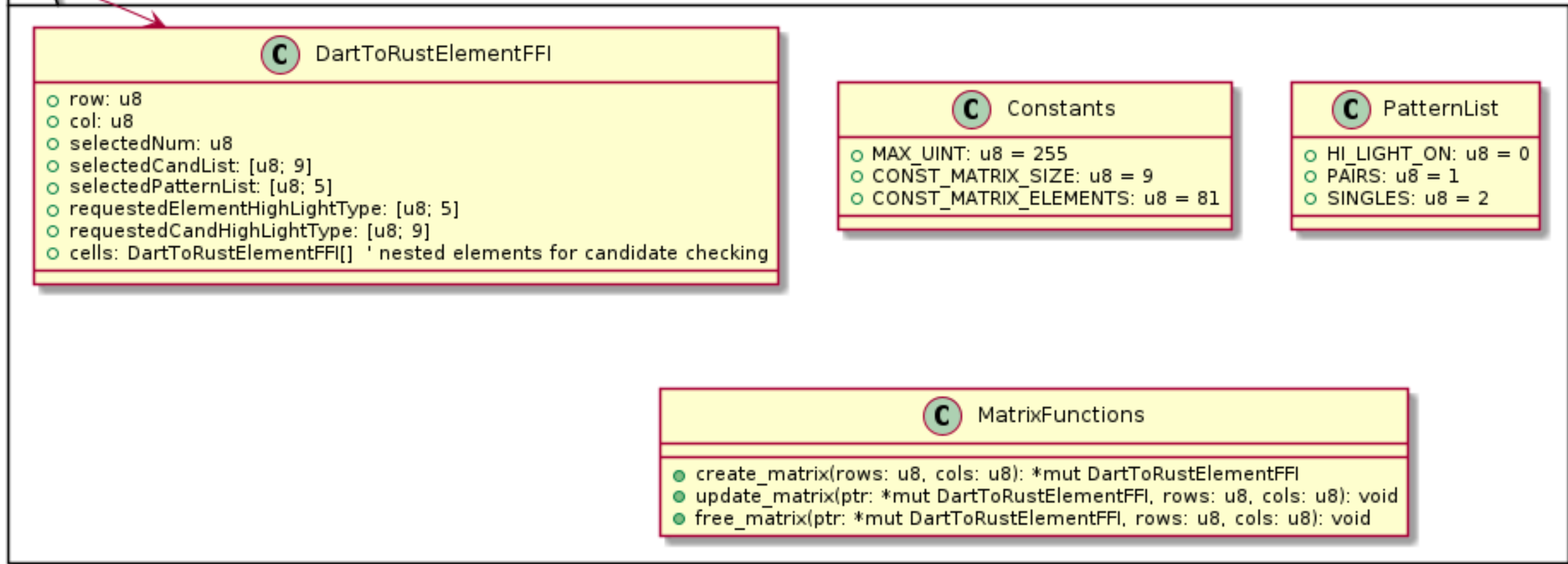
C ChangeNotifier



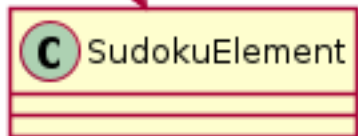
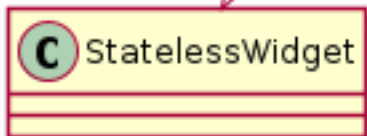
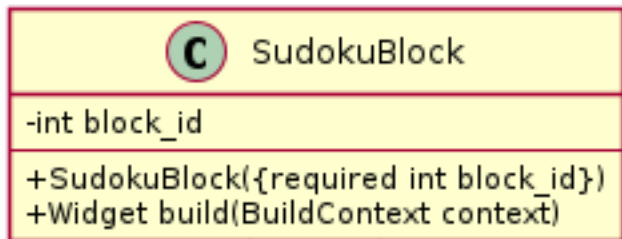
process_data



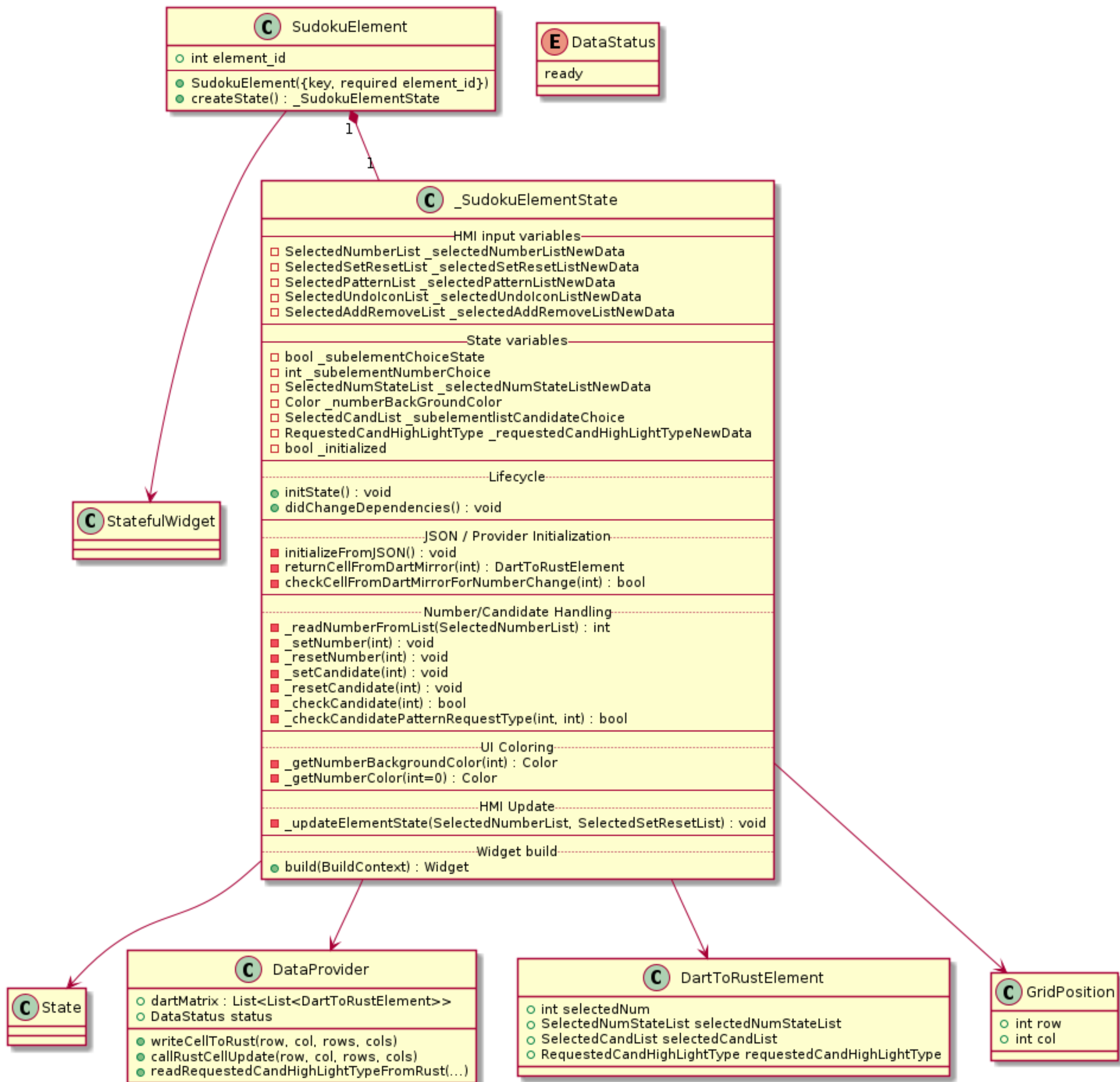
ffi



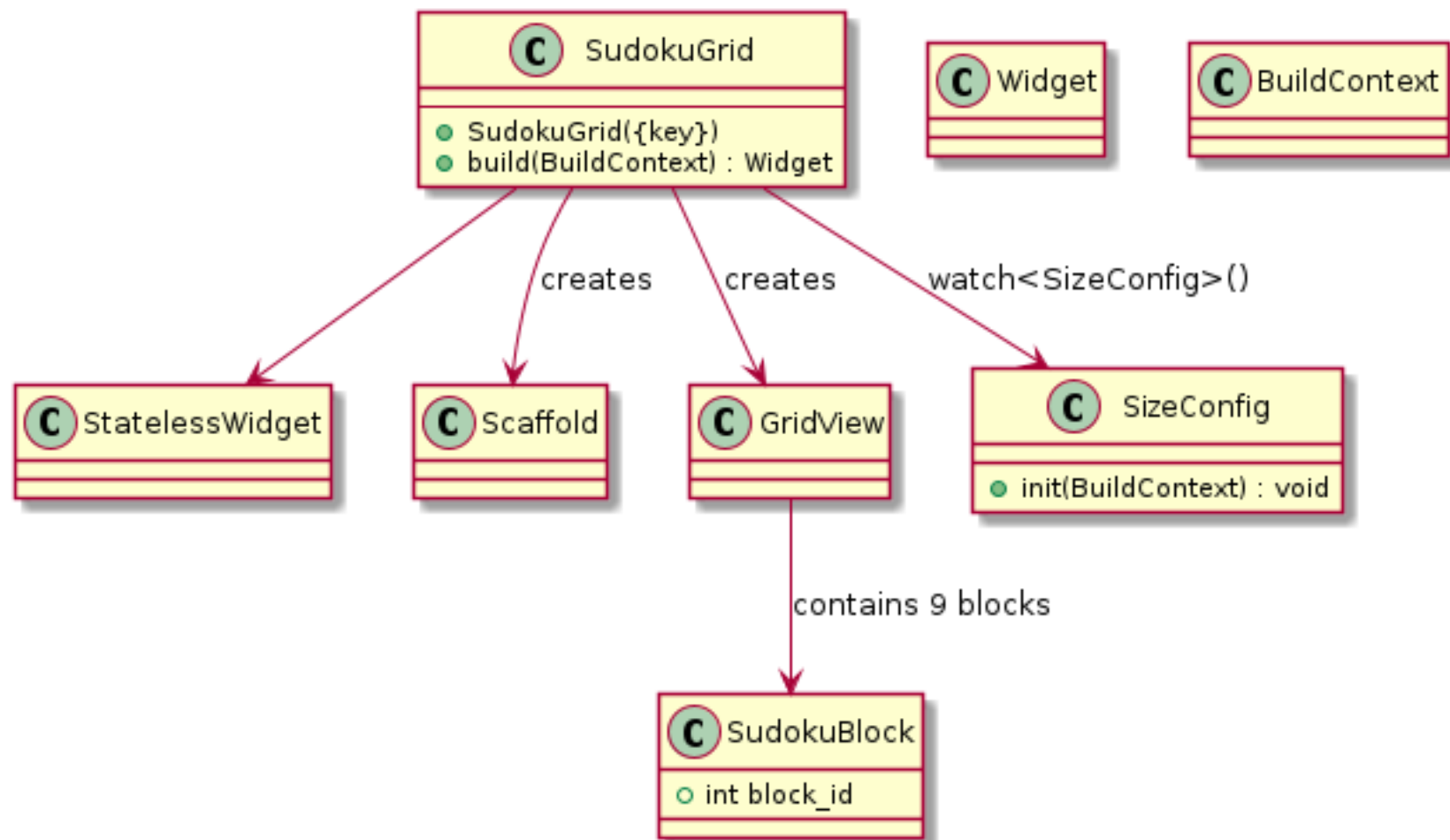
SudokuBlock Widget



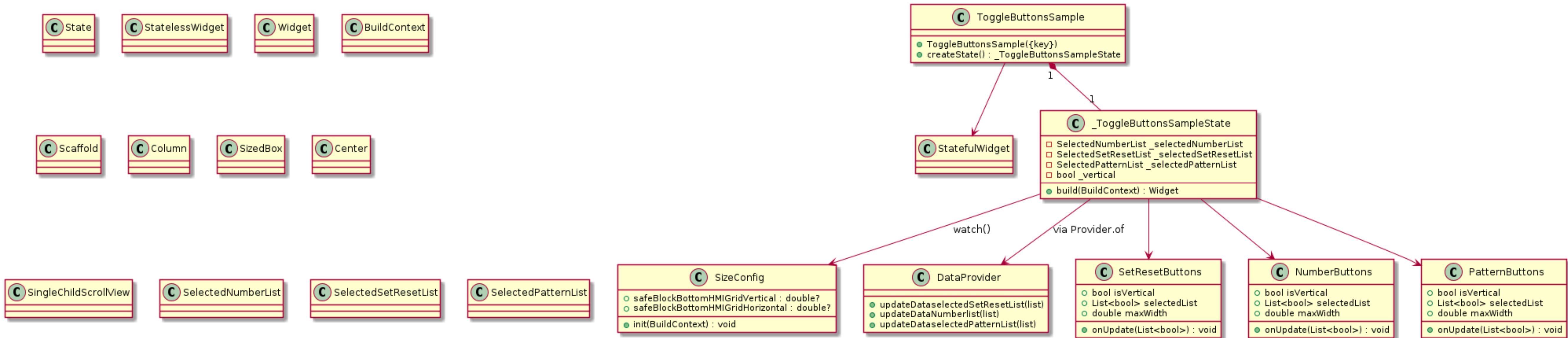
SudokuElement Class Diagram



SudokuGrid Class Diagram



ToggleButtonsSample Class Diagram



Unified Class Diagram- Sudoku UI Widgets

